

ThermalRadar

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	github.com/isemaster/ThermalRadar
	v0.1.10
	23 2026
	14 (4 , 4 , 4 , 2)

ThermalRadar. : SimulationManager, FlightSimulator, SignalProcessor, ThermalDetector, CirclingManager, VarioThermalDetector, WindEKF, WindStore, TrackReplayer, MainActivity . , DSP-, , .

1.

(SimulationManager + FlightSimulator)

CRITICAL 1.1. FlightSimulator: 0

FlightSimulator.start() altMsl = 0f (121). 0 500 . FlightStateMachine.update() ON_GROUND
FLYING, 0 " " . , altitude = 0, vario altMsl < 0, 0 (353), .

:

: altMsl = 500.0f (SimulationManager: INITIAL_ALT_MSL = 500.0f). FlightStateMachine .

CRITICAL 1.2. FlightSimulator: Quiet Period

FlightSimulator t=18 (231), (updateAccel) T_FREE_END=20 . SignalProcessor noiseFloor 100
(2) . (updateAccel), , . SimulationManager (lastPuffMs = -5000, 98), FlightSimulator .

:

: (2-3) FlightSimulator.updateAccel(), . t=22 , 2 .

HIGH 1.3. SimulationManager: noiseFloor

SimulationManager.generateAccel() NOISE_FLOOR_G * sin(noisePhase * 0.7 + 0.3), 0.003g . :
, , RMS , . SignalProcessor , , , noiseFloor . : noiseFloor , .

:

: : noiseX = NOISE_FLOOR_G * (2f * random.nextFloat() - 1f). Gaussian noise Box-Muller. .

HIGH 1.4. FlightSimulator:

FlightSimulator.updateAccel() circling=true 0.01-0.04g (417-422). TH_SUSPECT = 0.020g,
SNR > 3. NOISE_FLOOR_G = 0.003g 0.003-0.005g. 0.01g SNR = 0.01/0.005 = 2, SNR > 3.
ThermalDetector SUSPECT FlightSimulator.

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: FlightSimulator.updateAccel() 0.02-0.08g circling-, . (T_APPROACH).

2. (DSP)

: , . .

CRITICAL 2.1. Zero-Crossing:

SignalProcessor.process() (172-185) zero-crossing X Y: zcCountX zcCountY, : totalZc = zcCountX + zcCountY. = . X Y , totalZc = 2 * (). measuredFreq = totalZc / window_sec 2 . 2 , . , . , .

:

: A (): (X), zcCountY totalZc = zcCountX. measuredFreq = zcCountX / window_sec. B: () , 2 .

MEDIUM 2.2. noiseFloor: |bpX| + |bpY| RMS

SignalProcessor.process() (198-203) : sample = |bpOutX| + |bpOutY|. L1- (|X|+|Y|), RMS. sigma_x sigma_y L1- 0.8*(sigma_x + sigma_y), RMS = sqrt(sigma_x^2 + sigma_y^2). sigma_x = sigma_y = s: L1 = 1.6s, RMS = 1.41s. 13%, : noiseFloor , . , L1 RMS , .

:

: RMS: sample = sqrt(bpOutX*bpOutX + bpOutY*bpOutY). / ThermalDetector.

MEDIUM 2.3. :

SignalProcessor.process() (243-247) : dirX = (meanX >= 0) ? rmsX : -rmsX. : , . meanX , . , . , bandpass- DC- (). , , .

:

: " ": directionX = sum(bpX[i] * |bpX[i]|) / window, directionY = sum(bpY[i] * |bpY[i]|) / window. , , .

3.

README.md, VERSION_010.md, SIMULATION.md . , " " , .

HIGH 3.1. TH_SUSPECT: README 0.020g

ThermalDetector.java TH_SUSPECT = 0.020f (25). VERSION_010.md "TH_SUSPECT: 0.005g - 0.020g". README.md , . , TH_THERMAL = TH_SUSPECT = 0.020f, SUSPECT THERMAL. , , .

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: 1) README . 2) TH_SUSPECT TH_THERMAL (, 0.015g 0.025g) .

HIGH

3.2. System.currentTimeMillis() SystemClock.elapsedRealtime()

README " - dtMs elapsedRealtime". MainActivity.java System.currentTimeMillis(): 268, 483, 681, 800, 806, 939, 952, 1022, 1573, 1893, 1990, 2025, 2402, 2407. : lastThermalBeepMs (elapsedRealtime), ThermalBlip.bornMs (), GPS trail timestamps. (NTP,). , , , elapsedRealtime , currentTimeMillis .

:

: System.currentTimeMillis() SystemClock.elapsedRealtime(). System.currentTimeMillis() wall-clock (,).

MEDIUM

3.3. SIMULATION.md: v0.0.5, v0.1.10

SIMULATION.md ": v0.0.5" , . : SNR- (> 3), blip (ratio), TH_SUSPECT = 0.020g 0.005g. , SNR > 3, . SNR, .

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: SIMULATION.md: , SNR > 3, blip. , .

MEDIUM

3.4. WindStore: 10 , README 5

README.md: " (> 5) - ". WindStore.java: MAX_AGE_MS = 300_000 (5), CLEANUP_AGE_MS = 600_000 (10). cleanup() - (153). 5 getWindAt(), 10 . (), . , , getWindAt(), , .

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: README, CLEANUP_AGE_MS = MAX_AGE_MS = 300_000 .

4.

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MEDIUM

4.1. VarioThermalDetector: baseline

VarioThermalDetector.update() (61-91) baseline 30 . , baseline, . 30 baseline +1.5 / , effectiveThreshold = baseline + threshold . , , , , baseline .

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: baseline , thermalDetected = true. (). baseline vario < baseline ().

MEDIUM

4.2. CirclingManager:

CirclingManager.update() (295) EMA: windFromDeg += 0.3 * wdiff. EMA 0/360: windFromDeg = 350 newWindFrom = 10, wdiff = 20 (), . windFromDeg = 10 newWindFrom = 350, wdiff = -20, EMA 10 + 0.3*(-20) = 4, . , windFromDeg . , .

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:
: : sumSin += sin(windRad), sumCos += cos(windRad), avgWindFrom = atan2(sumSin,
sumCos). 0/360.
```

LOW

4.3. ThermalDetector: blip bornMs System.currentTimeMillis()

ThermalDetector.processSample() (254) ThermalBlip now = System.currentTimeMillis(). ThermalBlip.getBrightness() nowMs - bornMs. nowMs System.currentTimeMillis() elapsedRealtime(), (1.7). , , , , .

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:
: : bornMs (elapsedRealtime).
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LOW

4.4. TrackReplayer: IGC

TrackReplayer.loadFromIGC() (99-145) B- line.substring(7, 9), line.substring(25, 30) .. IGC, . StringIndexOutOfBoundsException catch (Exception). , (13:17:00 - 13:40:00) , IGC-. , IGC , .

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: 1) line.length() substring. 2) . 3) StringIndexOutOfBoundsException catch.
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5.

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1.1	0 FlightSimulator	CRIT	FlightSimulator	
1.2	Quiet Period FlightSimulator	CRIT	FlightSimulator	
1.3		HIGH	SimulationMana ger	
1.4	FlightSimulator	HIGH	FlightSimulator	
2.1	ZC:	CRIT	SignalProcessor	

2.2	noiseFloor: L1- RMS	MED	SignalProcessor	
2.3		MED	SignalProcessor	
3.1	SUSPECT/THERMAL	HIGH	ThermalDetector	
3.2	currentTimeMillis/elapsedRealtime	HIGH	MainActivity	
3.3	SIMULATION.md	MED	SIMULATION.md	
3.4	WindStore	MED	WindStore	
4.1	Baseline VarioThermalDetector	MED	VarioThermalDetector	
4.2	EMA ()	MED	CirclingManager	
4.3	bornMs:	LOW	ThermalDetector	
4.4	IGC	LOW	TrackReplayer	

∴

1 (30): FlightSimulator.start() - altMsl = 500f.

2 (30): Zero-Crossing SignalProcessor - .

3 (1): FlightSimulator.updateAccel() 0.02-0.08g.

4 (30): ∴ TH_SUSPECT = 0.015f, TH_THERMAL = 0.025f.

5 (2): System.currentTimeMillis() MainActivity, elapsedTime().

6 (1): VarioThermalDetector: baseline.

7 (1): EMA CirclingManager.

8 (30): SIMULATION.md README.md.

1-3 . 4-8 ∴ 6-7 .